

04: Categorical Variables + Intro to Quantitative Variables

Stat 120 | Fall 2025

Prof Amanda Luby

To get started:

1. Click the “code” button
2. Click “copy to clipboard” button
3. In maize, open a new rmarkdown file (New file → Rmarkdown → (name the file **04-activity.Rmd**))
4. Paste the code into the file that opened up and hit “save”

Once you have the file loaded into RStudio, you can follow the directions below.

The lab manual data **Pew** contains data from a January 2014 Pew Research Center survey about the Internet. We’ll consider two variables from this survey:

- **income**: the person’s yearly income, grouped into four categories
- **values**: an answer to the question asking if the Internet has been a “good thing”

0. Press the “play” button below to run the chunk of code. This loads the R *packages* that we need. Whenever you see this chunk in materials I give you, make sure to run it first!

```
library(tidyverse)
```

1. Import the data into R by running the following code chunk. Click on the dataset in the environment pane to pull up the data viewer. Can you see the “spreadsheet view”?

```
pew = read_csv("http://www.math.carleton.edu/Stat120/RLabManual/Pew.csv")
```

2. The code below computes the **frequency table** for the **values** variable. Add a “pipe” (`|>`) and another command to answer: what proportion of respondents said that the Internet *has* been a good thing?

```
table(pew$values)
```

```
bad good  
140  556
```

- Using `esquisse` or `ggplot`, make a bar plot of the counts of people who said the Internet has vs. has not been a good thing. Insert your code in the chunk below.
- Show the two-way table for the `values` and `income` variables. How many people from the survey had an income above \$150,000 a year and thought the Internet was a bad thing?

```
table(pew$income, pew$values)
```

	bad	good
0-30000	57	133
150000	5	64
30000-75000	50	206
75000-150000	28	153

- The table in part 3 shows the income categories in the wrong order (i.e., not in order of increasing income). You can fix this by changing the order of the categories of the `income` variable as below (see Appendix A.1 of the lab manual). Remake the table after reordering the categories.

```
pew$income <- factor(pew$income, levels = c("0-30000", "30000-75000",  
                                             "75000-150000", "150000"))
```

- Use `esquisse` or `ggplot` to make a *stacked barplot* of `income` and `values`.
- Your bar plot by default should show the *counts* of each combination. Copy and paste your bar plot code, and change `geom_bar()` to `geom_bar(position = "fill")`. Explain what this chunk of code did.

Checkpoint question

Summarize what you can conclude about the relationship between `values` and `income`

Stop Here and let Amanda know you've finished with the categorical EDA section

A third variable in the dataset is `age`. The next few questions walk through how to do basic EDA for a quantitative variable.

8. Create a histogram of the `age` variable using `esquisse` or `ggplot`
9. In the empty code chunk below, run `mean(pew$age)` and `median(pew$age)`. Now try `mode(pew$age)`. What happened?
10. Using your answers to the previous two questions, describe the distribution of `age`.